

Homework 2

Submission 3, Spring 2026

Sophie Kim

Problem 1

From the boxplots, the median number of plans increase consistently as time progresses. It seems like there is an excess number of plans, and although an increase of plan availability may suggest competition, there could be an overload of options to choose from for potential enrollees. Even after removing outliers, the median number of plans is more than 10 which could pose as a challenge for a layperson to compare effectively.

Distribution of Medicare Advantage Plan Counts by County (2014-2019)

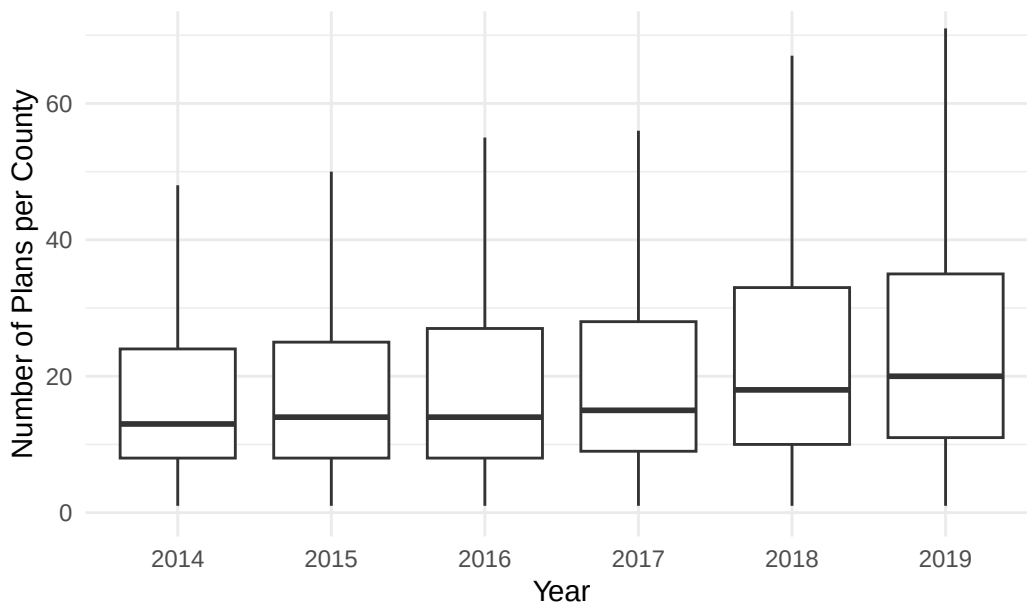


Figure 1: MA Plan Counts by County

Problem 2

After scaling the x and y axis for appropriate comparison of the 2014 and 2018 histogram, a subtle shift of the distribution to fewer plan bids (to the left) can be seen between 2014 to 2018. Additionally, the bid count substantially increased in 2018 compared to 2014, showing a larger number of lower bids being submitted.

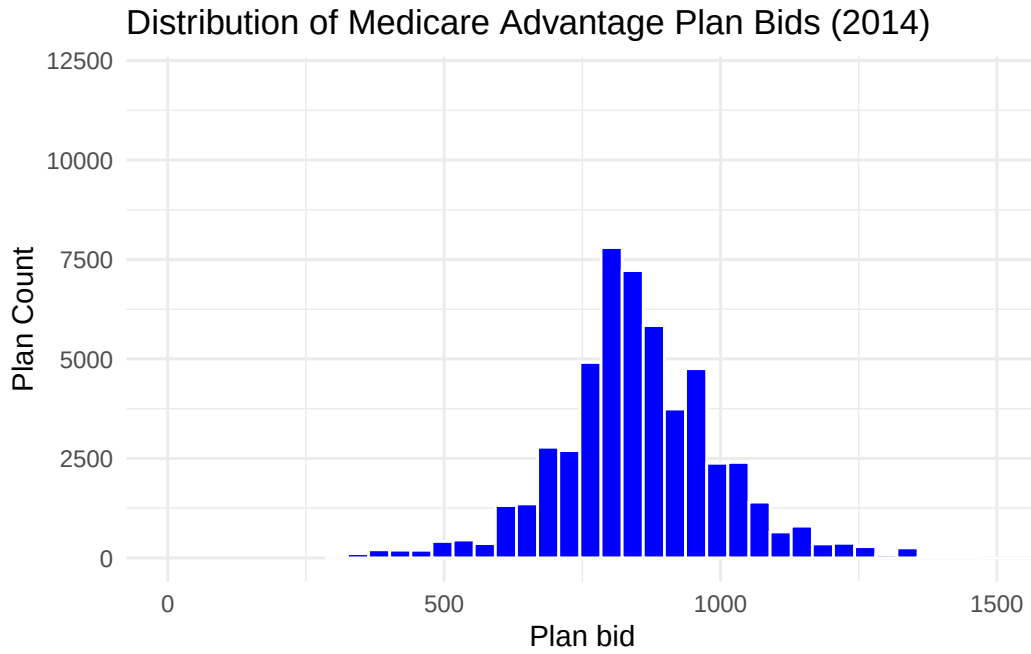


Figure 2: 2014 MA Plan Bids

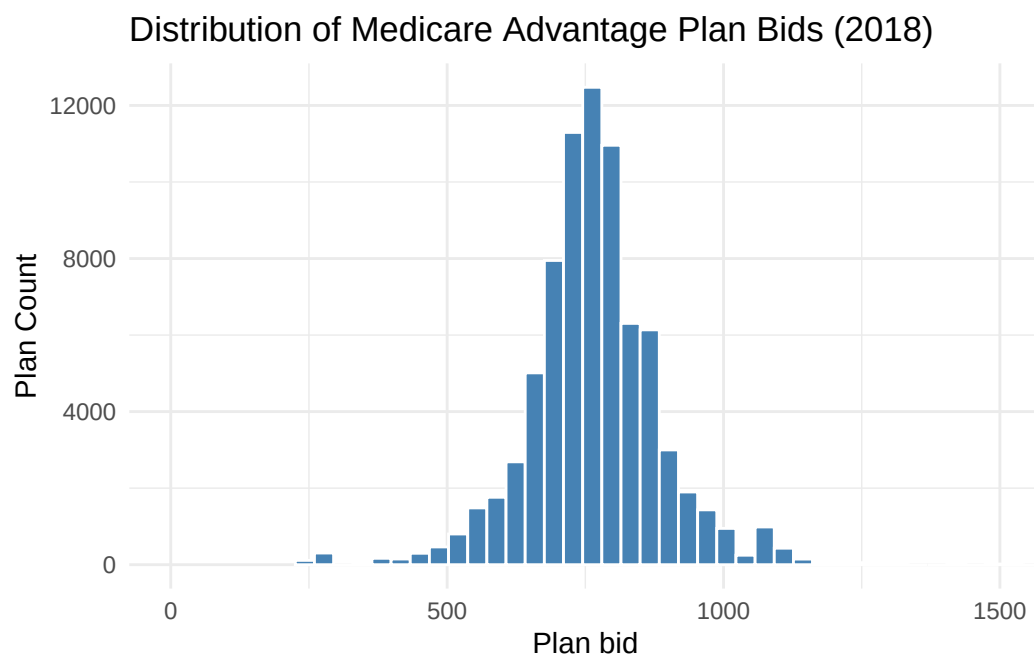


Figure 3: 2018 MA Plan Bids

Problem 3

HHI (0-1 scale used) decreases over time, which shows that the market's competitiveness increased from 2014 to 2019. Typically, since an HHI below 0.15 is considered competitive, the Medicare Advantage market can be seen as going from uncompetitive to competitive at around 2018.

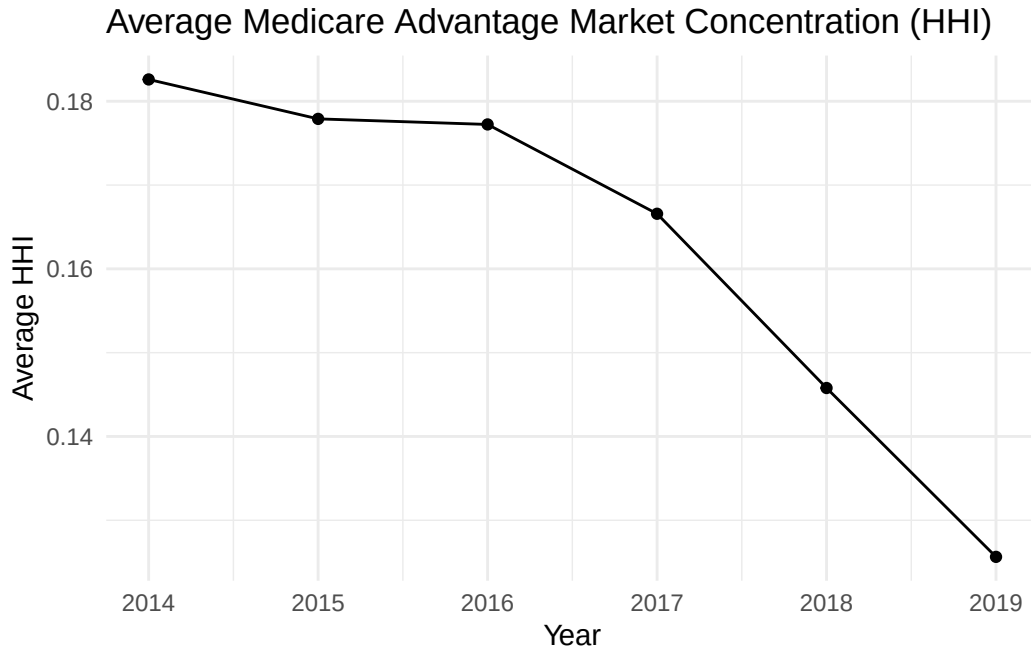


Figure 4: Average Medicare Advantage Market Concentration (HHI)

Problem 4

The average share of Medicare Advantage in the market (out of total of 1) over time increases consistently from 2014 to 2019.

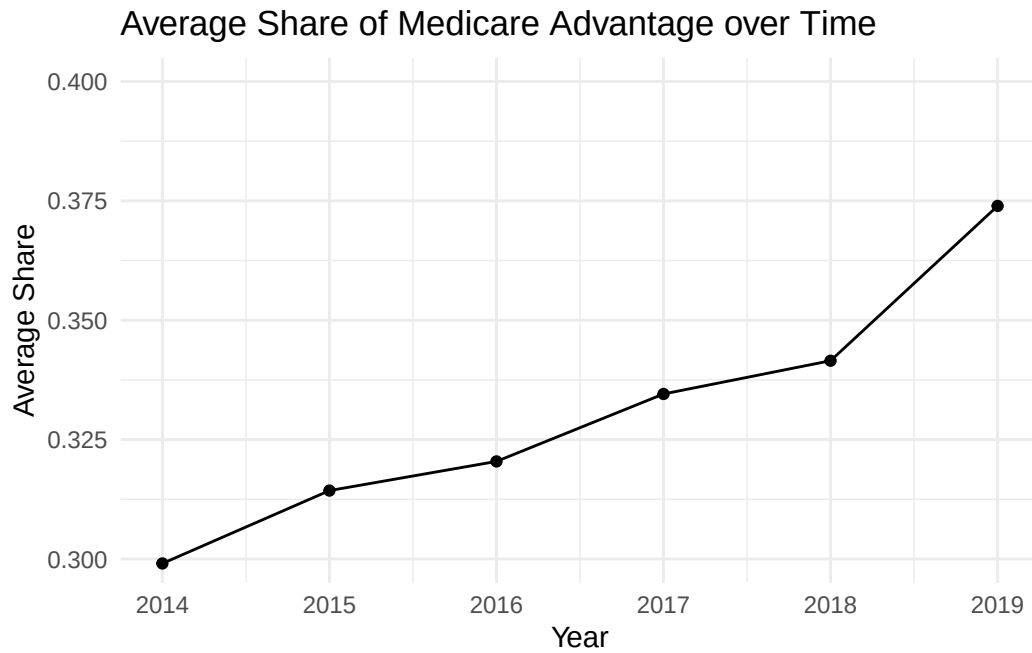


Figure 5: Average Share of Medicare Advantage over Time

Problem 5

Table 1: Average Bid by Market Type (2018)

Market type	Average bid	N markets
Competitive	782.8623	961
Uncompetitive	788.3218	915

Problem 6

Table 2: Mean Bids (2018)

FFS Quartile	Low HHI (Competitive)	High HHI (Uncompetitive)
Q1	788.81	814.56
Q2	784.27	780.50
Q3	775.19	765.66
Q4	783.97	784.83

Problem 7

Table 3: Average Treatment Estimators

	INV	MAH	IPW	OLS
High HHI	3.299	3.299	3.299	3.299

Problem 8

Nearest neighbor matching with inverse variance distance and with Mahalanobis distance have identical average treatment effect values as well as inverse propensity weighting and simple linear regression. This suggests that the estimated treatment effect is not sensitive to the choice of estimator.

Problem 9

The ATE value using inverse variance distance increases when using a continuous model for FFS and adding a covariant.

Table 4: Inverse Variance Distance

	INV Baseline	INV Alternate
High HHI	3.299	5.131

Problem 10

Working with this data allowed me to see how different estimators will give you identical values but the way that FFS costs are modeled (discrete vs continuous) can affect the average estimated treatment value. One thing that I learned was that it is crucial to know what each name of the row represents so that there is no issue missing a part of the dataset. I was surprised that the bid size between competitive and uncompetitive markets were quite close.